

Scalar Curvature Term $V''(z)$ in the Mexican Hat Potential Framework

The second derivative of the perturbed Mexican Hat potential, $V''(z)$, is constructed as:

$$V(z) = \lambda(\phi^2(z) - v^2)^2 + \delta \cdot J_{\text{injection}}(z)$$

Where:

- $\phi(z) \sim \mu(z)$ — a proxy scalar field derived from SN1a distance modulus data
- $J_{\text{injection}}(z)$ — redshift-dependent angular momentum injection profile derived from GWTC-3 events
- δ — a coupling parameter inferred through Bayesian sampling

$V''(z)$, the second derivative of this potential with respect to redshift, is computed numerically and interpreted as a scalar curvature contribution to cosmic acceleration. It serves as a physically grounded alternative to a cosmological constant (Λ), responding dynamically to gravitational wave-driven black hole mass accretion.